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Discrete Structure and Graph Theory

Discrete structure is a set of discrete elements on which certain operations are defined. Discrete implies noncontinuous and therefore discrete sets includes finite and countable sets but not uncountable sets.

A discrete structure is defined by a set of axioms.

Discrete structure or Discrete mathematics is the part of mathematics devoted to the study of discrete objects.

Discrete objects means that are - countable, separate, distinct.

Topics in discrete mathematics are - logic, functions & relations, counting, Graphs & trees etc.

Propositional logic :- A proposition or statement is a declarative sentence which is eighter true or false, but not both.

Propositional logic is concern with statements to which the truth values, "true" and "false", can be assigned.

Examples :-

1) There are seven days in a week 2) $2+2=5$

3) The Earth is flat 4) Equation $x^2+x+1=0$ has no real root.

5) It will rain tomorrow. 6) Four is even 7) $43>21$

Examples 1, 4, 6, 7 are true statements whereas statements 2, 3, are obviously false. In the case of example 5 although its true value cannot be predicted at this point of time, it can be definitely determined in the future. Hence it is also a statement

example, ⑧ $x+3=5$ is not a statement, since its truth value depends on the value of x. If $x=2$, the statement is true; If $x \neq 2$ the sentence is false. It is infact a propositional statement (function)

Examples, ⑨ Bring that Book! , ⑩ What a beautiful painting? , ⑪ This statement is false. 9, 10 & 11 are not declarative ~~statements~~ sentences. Hence these are not propositions

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A statement which does not contain any other statement is called simple statement example: It was raining on the last sunday
A statement (which can be split into simpler statements) which ~~ante~~ contains another statements is called a compound statement.
example. If you will commit a mistake, then you will pay for it.
logical operations (Symbolising statements)

① Negation:- (Not) [~(curl)/ ¬]
If 'p' is a statement, then the negation of p is ~p/¬p/ $\bar{p}$
If the truth value of 'p' is true, then truth value of ¬p is false.
p: I am going for a walk
¬p: I am not going for a walk
Truth table given on the right

p	¬p
T	F
F	T

② Conjunction (And) [Symbol: ∧]
If 'p' and 'q' are the statements, then "p and q" [p∧q] are called as the conjunction of 'p' and 'q'
p: The sun is shining
q: The birds are singing
p∧q: The sun is shining and the birds are singing

p	q	p∧q
T	T	T
T	F	F
F	T	F
F	F	F

③ Disjunction (Or) [symbol: V]
p: There is an error in the program.
q: The data is wrong
"p or q" (p V q): There is an error in the program or the data is wrong

p	q	p V q
T	T	T
T	F	T
F	T	T
F	F	F

④ Conditional/ Implication (if.... then)
p: Peter works hard (Hypothesis)
q: Peter will pass the exam (Conclusion)
p→q (p⊃q): If peter works hard, then he will pass the exam.
An implication is false only in one case: that is the Hypothesis is 'true' & the conclusion is 'false'

	p	q	p⊃q
①	T	T	T
②	T	F	F
③	F	T	T
④	F	F	T

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① → If peter works hard then he will pass the exam
② → If peter works hard then he will not pass the exam
③ → If peter does not work hard then he will pass the exam
④ → If peter does not work hard then he will not pass the exam.

The Converse of $p \to q$ is $q \to p$ / $(q \supset p)$
The Inverse of $p \to q$ is $\sim p \to \sim q$ / $(\neg p \supset \neg q)$
Contrapositive of $p \to q$ is $\sim q \to \sim p$ / $(\neg q \supset \neg p)$
Example:- "If you are good in Mathematics, then you are good in science"
Converse: $q \to p$: If you are good in science then you are good in Mathematics
Inverse: $\sim p \to \sim q$: If you are not good in math. then you are not good in science
Contrapositive: $\sim q \to \sim p$: If you are not good in sci then you are not good in Mathemat.

(5) Equivalance (Bidirectional)
Obtained by combining two simple statements by "If and only if"
If 'p' and 'q' are two statements then their equivalance is
$p \leftrightarrow q$ $(p \equiv q)$
$p \equiv q$ is a conjunction of $(p \supset q) \wedge (q \supset p)$ (and)

p	q	$p \supset q$	$q \supset p$	$p \equiv q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Logical Equivalence:-
Two statements are said to be logically equivalent if their equivalence is always true.
example ① $(p \supset q) \equiv (\sim p \vee q)$ show that
② Show that $(\sim P \wedge (\sim Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \leftrightarrow R$

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Tautology:- A statement form which is 'always true' for all substitution instances is called a Tautology

Example ① Prove that $[\sim(p \vee q)] \supset [\sim p \wedge \sim q]$ is a tautology.

Sol^n

p	q	$\sim p$	$\sim q$	$\sim p \wedge \sim q$	$p \vee q$	$\sim(p \vee q)$	$[\sim(p \vee q)] \supset [\sim p \wedge \sim q]$
T	T	F	F	F	T	F	T
T	F	F	T	F	T	F	T
F	T	T	F	F	T	F	T
F	F	T	T	T	F	T	T

② Prove that $[(p \to q) \wedge (q \to r)] \to (p \to r)$ is a tautology.

③ $\sim(q \wedge p) \vee (q \wedge \sim p)$ ~ ④ ~ ~~$p \vee \sim p$~~ $(p \vee \neg q) \wedge p$ ~

Contradiction:- A statement form which is 'always false' for all substitution instances is called contradiction.

example: Prove that $\sim[\{(p \vee q) \wedge \sim p\} \supset q]$ is a contradiction

p	q	$\sim p$	$p \vee q$	$(p \vee q) \wedge \sim p$	$[\{(p \vee q) \wedge \sim p\} \supset q]$	$\sim S$
T	T	F	T	F	T	F
T	F	F	T	F	T	F
F	T	T	T	T	T	F
F	F	T	F	F	T	F

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Laws of logic :-

1. Idempotent laws:
   (a) $p \vee p \equiv p$ (b) $p \wedge p \equiv p$
2. Commutative laws:-
   (a) $p \vee q \equiv q \vee p$ (b) $p \wedge q \equiv q \wedge p$
3. Associative laws:-
   (a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$ (b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
4. Identity laws:-
   (a) $p \vee \text{false} \equiv p$ (b) $p \wedge \text{false} = \text{false}$
   $p \vee \text{True} \equiv \text{True}$ $p \wedge \text{True} = p$
5. Distributive laws:-
   (a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
6. Absorption law
   (a) $p \wedge (p \vee q) \equiv p$ (b) $p \vee (p \wedge q) \equiv p$
7. Implication law:- $p \to q \equiv \sim p \vee q$
8. Complement laws / Inverse law
   (a) $p \vee \sim p \equiv \text{True}$ (b) $\sim(\sim p) = p$
   $p \wedge \sim p \equiv \text{False}$ $\sim T \equiv F$ & $\sim F \equiv T$
9. De-morgan's law.
   (a) $\sim(p \wedge q) \equiv \sim p \vee \sim q$ (b) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

logical equivalent

$\Rightarrow p \to q = \sim p \vee q$

$\Rightarrow p \leftrightarrow q = (p \to q) \wedge (q \to p)$
$= (\sim p \vee q) \wedge (\sim q \vee p)$

$\Rightarrow \sim p \vee \sim p = \sim p \;/\; p \vee p = p$

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Assignment :-

(a) Check if $(p \to q) \to [(\sim p \to q) \to q]$ is a tautology?

(b) Explain types of Quantifiers. Represent following sentences using Quantifiers

(i) Eulerian but not Hamiltonian

(ii) There is a student who can speak Hindi but doesnot know Marathi

(c) Prove Using mathematic Induction $1+5+9+\cdots+(4n-3) = n(2n-1)$

(d) write following statements in a symbolic form using quantifiers.

(i) All students have taken a course in mathematics

(ii) There is a girl student in a class who is also a sports person

(iii) Some students are intelligent, but not hardworking

(e) Using mathematical Induction, prove that the given equation is true for all positive integers $1\times3 + 3\times4 + 5\times6 + \cdots + (2n-1)\times 2n = n(n+1)(4n-1)/3$

(f) check whether $[(p \to q) \wedge \sim q] \to \sim p$ is a tautology.

(g) Prove by Mathematical Induction $2+5+8+\cdots+(3n-1) = n(3n+1)/2$

(h) Prove by " $1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$

(i) Prove that $8^n - 3^n$ is a multiple of 5 by Math. Ind^n $n \geq 1$

(j) Show that $n(n^2-1)$ is divisible by 24 where n is any ODD positive numb

(k) Prove that $7^n - 1$ is divisible by 6 for $n = 1, 2, 3, \ldots$

(l) " $6^{n+2} + 7^{2n+1}$ is divisible by 43

(m) Check tautology $(P \to Q) \leftrightarrow (\sim Q \to \sim P)$

(n) Obtain CNF and DNF of expression : $P \leftrightarrow (\sim p \vee \sim q)$

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Quantifiers:-

Sentences that contains a word indicating a quantity such as all, every, some, none, one are called propositions. The words indicating quantity are called quantifiers.

ex. All humans are mortals, Some people have pink hair, Nothing is perfect, There exists a white cow

Types of Quantifiers:- There are two types of quantifiers.

1. Universal quantifiers-

Universal quantifiers tells us that all (every, each) objects posses the property. It is denoted by '$\forall$' (an invert A). The symbol $\forall$ is read as "for all", "for each", for "every".

2. Existential quantifiers:-

The existential quantifiers tells us that "some" (at least one) objects posses the property. It is denoted by '$\exists$' (a backward E). The symbol $\exists$ is read as "for some", "there exist a" or "for at least one"

Symbolisation of Quantifiers (Predicate logic)

When we talking about some, all, none etc, we need reference set which we call "Universe of discourse". If we ~~talk~~ talking about "some students", the universe of discourse is the set of all students. We shall denote a member of the set (i.e. of the universe of discourse) by x (or y or z). The property possessed by x is denoted by $P(x)$ & $P(x)$ is called a predicate. The predicate $P(x)$ states the property the object x has. (Predicate defines relation between objects)

For example, $P(x)$: x is yellow, $P(x)$: x is married are predicates. $P(x)$ is not a proposition but it is just an expression. As such $P(x)$ is not true or false. It becomes a proposition when x is assigned a value, some, or all. A proposition can be true or false.

For example:-

$(\forall x)\, P(x)$ ~~stands for~~ is interpreted as "for all values of x, the assertion $P(x)$ is true." is a statement in which the variable 'x' is said to universally quantified.

The meaning of $\forall$ is "for all" or "for every" or "for each"
If $p(x)$ is true for every possible value of x, then $\forall x\ p(x)$ is true
Otherwise $\forall x\ p(x)$ is false.
Suppose for the predicate $p(x)$, $\forall x\ p(x)$ is false, but there exists atleast one value of x for which $p(x)$ is true, then the variable x is said to be existential quantified. We denote the words "there exists" by the symbol $\exists$.
Then the notation $\exists x\ p(x)$ means "there exists a value of x (in the universe of discourse) for which $p(x)$ is true"
Example:- If the Universe of discourse is $A = \{1,2,3,4,5,6\}$, determine the truth value of the following
(i) $(\exists x)(x+4=12)$ (ii) $(\forall x)(x+4<12)$
(iii) $(\exists x)(x+2<6)$ (iv) $(\forall x)(x+2\leq 7)$
Sol^n:- (i) Let $p(x)$: $x+4=12$ $\therefore x=8$, the proposition is true, But $8\notin A$
There exist a value of x for which $p(x)$ is true. But this statement is false.
(ii) Let $p(x)$: $x+4<12$ [$\forall x\ p(x)$: for all x, $p(x)$ is true]
since for all $x\in A$, the proposition is true
(iii) $(\exists x)(x+2\leq 6)$: There exist a value of x for which $p(x)$ is true
Let $p(x)$: $x+2\leq 6$
for some value of $x=1,2,3$ the proposition is true
$\therefore (\exists x)(x+2\leq 6)$ is true.
(iv) $(\forall x)(x+2\leq 7)$ $\therefore$ for all value of x, the $p(x)$ is true
$p(x)$: $x+2\leq 7$
for $x=1,2,3,4,5$, $p(x)$ is true but not for $x=6$.
$\therefore$ Proposition $(\forall x)(x+2\leq 7)$ is false for all

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Normal forms:-

In case of large number of component statements truth table method is not a very convenient method of deciding whether the component statement is a tautology or a contradiction. The alternate procedure is known as reduction to normal forms. There are two such normal forms.

1) Disjunctive Normal form (DNF):-

A statement form which is a "disjunction of fundamental conjunctions" is called disjunctive normal form. A conjunction of statements and/or the negations of conjunctions of the statements is called a 'fundamental conjunction'. (e.g. p, $\sim p$, $p \wedge q$, $\sim p \wedge q$, $p \wedge \sim q$; $p \wedge \sim q$ etc. are fundamental conjunctions)

The following are some of the disjunctive normal forms.

(i) $(p \wedge q) \vee \sim q$ (ii) $(p \wedge q) \vee (p \wedge r)$ (iii) $(p \wedge \sim q) \vee (p \wedge \sim r)$

(iv) $(p \wedge q) \vee (q \wedge r)$.

But $p \wedge (q \vee r)$ is not a disjunctive normal form as it is not a disjunction of fundamental conjunctions.

2) Conjunctive Normal form (CNF)

A statement form which is a "conjunction of fundamental disjunctions" is called conjunctive normal form. A disjunction of statements and/or the negation of disjunctions of the statements is called a fundamental disjunction. For example, p, $\sim p$, $p \vee q$, $\sim p \vee q$, $p \vee \sim q$, $\sim p \vee \sim q$ etc are fundamental disjunctions.

The following are some of conjunctive normal forms

(i) $p \wedge r$ (ii) $\sim p \wedge (p \vee r)$ (iii) $(p \vee \sim q) \wedge (p \vee \sim r)$ (iv) $(p \vee q) \wedge (q \vee r)$

Example:- find the conjunctive normal form of $\sim[(p \vee \sim q) \wedge \sim r]$

Soln:- We have, $\sim[(p \vee \sim q) \wedge \sim r] = \sim(p \vee \sim q) \vee \sim\sim r$ ... [De Morg]

$= (\sim p \wedge \sim\sim q) \vee \sim\sim r$ ... [De Morgen]

$= (\sim p \wedge q) \vee r$ ...

$= r \vee (\sim p \wedge q)$ ... [commutative]

$= (r \vee \sim p) \wedge (r \vee q)$ .. [Distributive]

Example:-

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Examples of Quantifiers :-

→ $\forall y \exists x\ P(x,y)$: for each All value of 'y', there exists an 'x' such that P(x,y) is t

→ $\exists x\ \exists y\ P(x,y)$: There exists a value of 'x' and a value of 'y' such that P(x,y) is tru

→ ~~$\exists y\ \forall x\ P(x,y)$~~: There exists an 'y' $\exists x\ \forall y\ P(x,y)$ is false.

→ $\forall x\ \forall y\ P(x,y)$: for all values of 'x' and 'y', P(x,y) is true.

→ Example:- There is a student in this class, who is not familiar with 'C' programi

Symbolically, $\exists x(\sim P(x))$ ..... There exists 'x', P(x) is false

P(x) : familiar with C programing

→ Example:- All students have taken a course in communication skill

Ans:- $\forall x\ P(x)$ ... P(x): students ~~x~~ has taken a course in comm^n skill

→ Example:- There is a girl student in the class who is also a sports pers

Ans:- P(x) : students

Q(x) : girl

R(x) : sport person

$\therefore \exists x\ [P(x) \wedge Q(x) \wedge R(x)]$

→ Some students are intelligent, but not hardworking

P(x) : intelligent students Q(x) : Hardworking students

$\exists x\ [P(x) \wedge \sim Q(x)]$

→ ① All students go to party ② Some students eat too much

③ Some students who eat too much go to the party.

Ans :- S(x) : students, D(x) : students eat too much

P(x) : students goes to party

① $\forall x\ [S(x) \rightarrow P(x)]$ ② $\exists x\ [S(x) \wedge D(x)]$

③ $\exists x\ [D(x) \wedge P(x)]$

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Mathematical Induction

① Example:- Prove by induction: $1+2+3+\dots+n = \frac{n(n+1)}{2}$ for all natural number values of n.

Soln :- let $p(n)$ be the statement :- $p(n) = \underbrace{1+2+3+\dots+n}_{LHS} = \underbrace{\frac{n(n+1)}{2}}_{RHS}$

step-I] for $n=1$, $p(1) = \underset{LHS}{1} = \underset{RHS}{\frac{1(1+1)}{2}} = 1$

Hence $p(1)$ is true

for $n=2$, $p(2) = 1+2 = \frac{2(2+1)}{2} = 3$

Hence $P(2)$ is true.

step II] Assume $P(k)$ is true.

$P(k) = 1+2+3+\dots+k = \frac{k(k+1)}{2}$ ---- (i)

prove $P(k+1)$ is also true.

$P(k+1)$: $1+2+3+\dots+k+(k+1) = \frac{(k+1)(k+2)}{2}$ .... (ii)

Put eqn (i) in LHS of eqn (ii)

$P(k+1)$: LHS $= 1+2+3+\dots+k+(k+1)$

$= \frac{k(k+1)}{2} + k+1 = \frac{k^2+k+2k+2}{2} = \frac{(k+1)(k+2)}{2}$ = RHS of eqn (ii)

Hence assuming $p(k)$ is true, $P(k+1)$ is also true.

Therefore $P(n)$ is true ~~of~~ for all natural number values.

② Prove by mathematical induction: $1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$

soln:- let $p(n) = 1^2+2^2+3^2+\dots+n^2 = \frac{n(n+1)(2n+1)}{6}$

step I] for $n=1$; $P(1) = 1^2 = \frac{1(1+1)(2+1)}{6} = 1$

for $n=2$; $P(2) = 1^2+2^2 = \frac{2(2+1)(4+1)}{6}$

$\therefore 5 = \frac{2(3)(5)}{6} = 5$

Hence $P(1)$ & $P(2)$ are true.

step II] Assume $P(k)$ is true.

$P(k)$: $1^2+2^2+3^2+\dots+k^2 = \frac{k(k+1)(2k+1)}{6}$ ..... (i)

prove $p(k+1)$ is also true.

$p(k+1)$: $1^2+2^2+3^2+\dots+k^2+(k+1)^2 = \frac{(k+1)[k+1+1][2k+2+1]}{6}$

$= \frac{(k+1)(k+2)(2k+3)}{6}$ .... (ii)

Put eqn (i) in LHS of eqn (ii)

$P(k+1)$: LHS $= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 = (k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right]$

$= \frac{(k+1)}{6}\left[k(2k+1) + 6(k+1)\right] = \frac{k+1}{6}\left[2k^2+k+6k+6\right]$

with any function ...

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$= \frac{(k+1)}{6}\left[2k^2+7k+6\right] = \frac{(k+1)}{6}\left(2k^2+4k+3k+6\right)$

$= \frac{(k+1)}{6}\left(\frac{2k(k+2)+3(k+2)}{1}\right) = \frac{(k+1)(k+2)(2k+3)}{6}$

$=$ RHS of eqn (ii)

Hence assuming $P(k)$ is true, $P(k+1)$ is also true

Therefore $P(n)$ ~~true~~ is true by mathematical Induction

(3) show that n^3+2n is divisible by 3 for all $n \geq 1$

→ Since $n \geq 1$ let $n=1$ & $P(n) = n^3+2n$

$P(1) = 1^2+2 = 3$ ∴ 3 is divisible by 3

Assume that for $n=k$ it is true

that is, $P(k) = k^3+2k$ (is divisible by 3) ..... (i)

We will prove it for $n=k+1$

$\therefore P(k+1) = (k+1)^3+2(k+1) = [k^3+3k^2(1)+3(1)^2k+(1)^3]+2k+2$

$= k^3+3k^2+3k+2k+1+2 = \cancel{k^3+3k^2+10k+3}$

$= (k^3+2k)+3k^2+3k+3 = (k^3+2k)+3(k^2+k+1)$

Since k^3+2k is divisible by 3 as per eqn (i) and $3(k^2+k+1)$ is also divisible by 3 therefore we can say n^3+2n is divisible by 3.

(4) Proove using mathematical Induction that $1+5+9+\cdots+(4n-3) = n(2n-1)$

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(5) show that $1^2+3^2+5^2+\cdots+(2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$